DEVELOPMENT INTEGRATION HANDOFF

VRS Express to VRS AI

Strict V5 and strict V8 development integration reference

USE IN CONTROLLED DEVELOPMENT ONLY

This package contains no live URL, credential, or customer data. Substitute only the four approved development placeholders and use development-safe test data. A 202 admission response is not proof that a phone call has started or completed.

Item	Value
Platforms	VRS Express and VRS AI
Interface	Strict V5 and strict V8 on POST /vrs/authorize
Environment	Development only until both acceptance directions pass
Bundle	This PDF, editable DOCX, and vrs-express-vrs-ai-bridge-openapi.yaml

This is an in-place revision of the existing bridge, not a replacement system. The YAML file in this package is the machine-readable OpenAPI 3.1 contract; this document is the operating guide for the two development test directions.

1. Platform responsibilities

Platform	Responsibility
VRS Express	Authenticate authorize intake; create the normal pending ticket; hold for human approval; dispatch the same contract version; verify result signatures; apply ticket and SMS effects once; and forward the result.
VRS AI	Simulate the upstream sender for inbound testing; receive approved requests for outbound testing; own call admission, rate limits, call execution, recordings, callback production, and callback retries.

MANUAL-FLOW ISOLATION

Manual and PWA tickets with no bridge record never call VRS AI.

The current automated bridge is for American Home Shield authorization. VRS Express normalizes partner_id case, spaces, and hyphens and accepts ahs, american_home_shield, or N/A; missing V5 partner_id defaults to AHS. Other partner values are rejected before ticket creation.

Approved development placeholders

Placeholder	Required development value

Placeholder	Required development value
<VRS_EXPRESS_DEV_BASE_URL>	Public VRS Express development origin, without /vrs/authorize
<VRS_AI_DEV_BASE_URL>	Public VRS AI development origin, without /vrs/authorize
<VRS_EXPRESS_DEV_CALLBACK_URL>	Exact VRS Express development callback ending /api/integrations/vrs-ai/v1/webhook
<UPSTREAM_DEV_CALLBACK_URL>	Exact clean HTTPS upstream development webhook already permitted by VRS Express intake

Development data prerequisites

- Replace each sample tech_id with a provisioned active VRS Express DEV technician mapped to a valid four-digit district. Use the same ID in comments[].tech_id.
- Replace each sample service_order with a unique eight-digit DEV value. The literal sample identifiers are format-safe only and must not be sent unchanged.

2. Exact two-direction development flow

Direction A: VRS AI simulator to VRS Express

1. VRS AI's simulator sends POST <VRS_EXPRESS_DEV_BASE_URL>/vrs/authorize.
2. It authenticates with the development VRS Express intake X-API-Key.
3. It sends one complete strict V5 or V8 body with callback_url set to <UPSTREAM_DEV_CALLBACK_URL>.
4. VRS Express returns 202 with its public call ID after creating one pending human-review ticket. This 202 means pending human review and does not call VRS AI.
5. Before human approval, the expected VRS AI receive count is zero.

Direction B: VRS Express to VRS AI

1. A human reviews and approves the development ticket.
2. VRS Express sends the same contract version to <VRS_AI_DEV_BASE_URL>/vrs/authorize using the separate VRS Express-to-VRS AI development X-API-Key.
3. VRS Express preserves the business payload and replaces only callback_url with <VRS_EXPRESS_DEV_CALLBACK_URL> and callback_secret with its per-attempt callback secret.
4. A VRS AI 202 means asynchronous admission. It does not prove that dialing or any phone call has started or completed.
5. VRS AI returns its provider call_id, then sends signed events to the exact callback URL received in the request.

The two X-API-Key values are separate credentials and are exchanged outside this handoff.

3. Strict V5/V8 selection

ONE ENDPOINT, TWO STRICT SHAPES

Both versions use the same POST /vrs/authorize route and reject additional root, part, and comment properties.

- If any one of appliance_color, contract_number, purchase_year, or symptom exists, the entire request is strict V8.
- Otherwise the entire request is strict V5.
- A partial V8 request fails V8 validation; it never falls through to V5.
- VRS Express dispatches the same contract version received and never up-converts V5.
- Never emit contractVersion, contract_version, cta_no, end_level, or diagnostic_photos.

4. Copyable V5 development payload

V5 requires the core fields below. appliance_grade, partner_id, appliance_photos, and comments are optional, but this complete example includes them.

```
{
  "service_order": "13538277",
  "tech_id": "replace-with-dev-tech-id",
  "tech_phone": "+12145550199",
  "tech_name": "Example Technician",
  "reason_for_call": "authorization",
  "callback_url": "<UPSTREAM_DEV_CALLBACK_URL>",
  "callback_secret": "example-callback-secret-not-for-live-use",
  "parts": [
    {
      "part_id": "WPW10503278",
      "part_description": "Electronic Control Board",
      "sell_price": 245,
      "part_function": "Controls wash cycle timing and motor speed."
    }
  ],
  "customer_name": "Example Customer",
  "customer_address": "100 Example Street, Example City, NC 27000",
  "service_address": "100 Example Street, Example City, NC 27000",
  "brand": "Whirlpool",
  "model_number": "WTW5000DW1",
  "serial_number": "C31290822",
  "appliance_type": "Washer",
  "appliance_age": 4,
  "labor_amount": 120,
  "tax_amount": 15.6,
  "part_price": 245,
  "appliance_location": "laundry room",
  "failure_cause": "normal-wear",
  "tech_recommendation": "repair",
  "repair_replace_rationale": "The isolated board failure is repairable.",
  "appliance_grade": "mid",
  "partner_id": "ahs",
  "appliance_photos": [],
  "comments": [
    {
      "tech_id": "replace-with-dev-tech-id",
      "work_item": "failure_diagnosis",
      "text": "Diagnostic code F7E1 was confirmed.",
      "timestamp": "2026-08-11T18:00:00Z"
    }
  ]
}
```

5. Copyable V8 development payload

V8 requires every V5 key, including the four fields optional in V5, plus appliance_color, contract_number, purchase_year, and symptom.


```
{
  "service_order": "13538278",
  "tech_id": "replace-with-dev-tech-id",
  "tech_phone": "+12145550199",
  "tech_name": "Example Technician",
  "reason_for_call": "authorization",
  "callback_url": "<UPSTREAM_DEV_CALLBACK_URL>",
  "callback_secret": "example-callback-secret-not-for-live-use",
  "parts": [
    {
      "part_id": "WPW10503278",
      "part_description": "Electronic Control Board",
      "sell_price": 245,
      "part_function": "Controls wash cycle timing and motor speed."
    }
  ],
  "customer_name": "Example Customer",
  "customer_address": "100 Example Street, Example City, NC 27000",
  "service_address": "100 Example Street, Example City, NC 27000",
  "brand": "Whirlpool",
  "model_number": "WTW5000DW1",
  "serial_number": "C31290822",
  "appliance_type": "Washer",
  "appliance_age": 4,
  "labor_amount": 120,
  "tax_amount": 15.6,
  "part_price": 245,
  "appliance_location": "laundry room",
  "failure_cause": "normal-wear",
  "tech_recommendation": "repair",
  "repair_replace_rationale": "The isolated board failure is repairable.",
  "appliance_grade": "mid",
  "partner_id": "American Home Shield",
  "appliance_photos": [],
  "comments": [
    {
      "tech_id": "replace-with-dev-tech-id",
      "work_item": "failure_diagnosis",
      "text": "Diagnostic code F7E1 was confirmed.",
      "timestamp": "2026-08-11T18:00:00Z"
    }
  ],
  "appliance_color": "stainless steel",
  "contract_number": "1234567890",
  "purchase_year": 2022,
  "symptom": "Washer does not drain"
}
```


V8 placeholder values

- tech_phone: E.164 or N/A.
- appliance_grade: entry, mid, top, or N/A.
- appliance_color, contract_number, symptom, and partner_id: nonblank; N/A is permitted.
- purchase_year: a positive integer with no contract upper limit, TBD, or N/A.
- tech_name, customer_name, parts[].part_function, and repair_replace_rationale may be the empty string.
- appliance_photos and comments may be empty arrays but must be present.

6. Exact field constraints

Unknown root, part, and comment keys are rejected.

Field	V5	V8
service_order	Exactly 8 digits	Exactly 8 digits
tech_id	Nonblank, max 50	Nonblank, max 50
tech_phone	E.164	E.164 or N/A
tech_name	Nonblank, max 255	String, max 255; empty allowed
reason_for_call	Literal authorization	Literal authorization
callback_url	Direction-specific DEV callback URL	Same
callback_secret	16-512 characters	16-512 characters
parts	At least 1; no contract maximum	At least 1; no contract maximum
customer_name	Nonblank, max 255	String, max 255; empty allowed
customer_address; service_address	Nonblank, max 1000 each	Same
brand; model_number; serial_number; appliance_type	Nonblank, max 100 each	Same
appliance_age	Number, 0-100	Number, 0-100
labor_amount; tax_amount; part_price	0-1,000,000; max 2 decimals	Same
appliance_location	Nonblank, max 100	Nonblank, max 100
failure_cause	normal-wear, power-surge, impact, installation-error, abuse, age, or other	Same
tech_recommendation	repair or replace	Same
repair_replace_rationale	Nonblank; no contract maximum	String; empty allowed; no contract maximum
appliance_grade	Optional: entry, mid, top	Required: entry, mid, top, N/A
partner_id	Optional nonblank string, max 50	Required nonblank string, max 50
appliance_photos	Optional; max 10 absolute URLs	Required; max 10 absolute URLs
comments	Optional; max 20	Required; max 20
appliance_color	Not allowed	Required nonblank string, max 40
contract_number	Not allowed	Required nonblank string; no contract maximum
purchase_year	Not allowed	Positive integer with no upper limit, TBD, or N/A
symptom	Not allowed	Required nonblank string; no contract maximum

Strict parts[] items

Field	Constraint
part_id	Nonblank, max 100
part_description	Nonblank, max 500
sell_price	0-1,000,000; at most 2 decimals
part_function	Max 200; V5 nonblank, V8 empty allowed

Strict comments[] items

Field	Constraint
tech_id	Nonblank, max 50
work_item	failure_diagnosis, tech_recommendation, appliance_features, customer_history, tech_observation, or other
text	Nonblank, max 2000
timestamp	ISO-8601 date-time with offset

7. Controlled V8 enrichment and photos

After approval, VRS Express may fill only a documented placeholder in an already-existing V8 key from authoritative data for that exact field. The implemented candidates are tech_phone, tech_name, and an empty appliance_photos array. It never overwrites a populated value, appends a missing key, adds cta_no, or changes contract version.

PHOTO URL INTEGRITY

Sender-provided TINA/signed photo URLs pass through byte-for-byte and require no VRS Express login. VRS Express does not convert authenticated internal /objects/ paths into outbound sender photo URLs.

8. Authorize responses and identifiers

Accepted response

```
{
  "call_id": "vrs_call_456",
  "service_order": "13538278",
  "state": "queued",
  "created_at": "2026-08-11T18:01:00Z"
}
```

For Direction A, call_id is the public VRS Express call ID. For Direction B, it is the provider VRS AI call ID. VRS Express stores both and never exposes the provider ID in its upstream admission response.

Status	Meaning
400	Strict validation failed
401	Missing or invalid X-API-Key
409	DUPLICATE_CALL includes existing_call_id; NOT_A_TECHNICIAN, SUBMISSION_ALREADY_EXISTS, and SUBMISSION_REJECTED when the captured ticket status maps to a conflict return { error, message }.
422	Unsupported partner code UNSUPPORTED_PARTNER includes partner_id; UNKNOWN_TECHNICIAN, INACTIVE_TECHNICIAN, INVALID_TECHNICIAN_DISTRICT, and non-conflict SUBMISSION_REJECTED return { error, message }.
429	Admission limit reached; body includes retry_after_seconds
500, 502, 503, timeout	Admission was not proven

VRS AI owns rate limits and its 429 policy. VRS Express does not add a second contract rate schedule.

9. VRS AI result callback

VRS AI sends POST <VRS_EXPRESS_DEV_CALLBACK_URL> with the following headers:

```
Content-Type: application/json
X-VRS-Signature: sha256=64-lowercase-hex-digest
X-VRS-Event-Id: evt_001
X-VRS-Timestamp: 1786471260000
```

All events require version: 1, eventId max 100, callId max 100, exact 8-digit soNo, an ISO-8601 occurredAt, and one of the five shapes below.

approved

```
{
  "version": 1,
  "eventId": "evt_approved_001",
  "callId": "vrs_ai_call_456",
  "soNo": "13538278",
  "occurredAt": "2026-08-11T18:04:00Z",
  "state": "approved",
  "authNumber": "AHS-001",
  "authorizedAmount": 365.6,
  "durationSec": 187
}
```

partial

```
{
  "version": 1,
  "eventId": "evt_partial_001",
  "callId": "vrs_ai_call_456",
  "soNo": "13538278",
  "occurredAt": "2026-08-11T18:04:00Z",
  "state": "partial",
  "authNumber": "AHS-002",
  "authorizedAmount": 250,
  "requestedAmount": 380.6,
  "denyReason": "Labor exceeded the coverage limit",
  "durationSec": 214
}
```

denied

```
{
  "version": 1,
  "eventId": "evt_denied_001",
  "callId": "vrs_ai_call_456",
  "soNo": "13538278",
  "occurredAt": "2026-08-11T18:04:00Z",
  "state": "denied",
  "denyReason": "Repair cost exceeds the authorization limit",
  "durationSec": 142
}
```

failed

```
{
  "version": 1,
  "eventId": "evt_failed_001",
  "callId": "vrs_ai_call_456",
  "soNo": "13538278",
  "occurredAt": "2026-08-11T18:04:00Z",
  "state": "failed",
  "reason": "busy",
  "retryable": true
}
```

requires_info

```
{
  "version": 1,
  "eventId": "evt_info_001",
  "callId": "vrs_ai_call_456",
  "soNo": "13538278",
  "occurredAt": "2026-08-11T18:04:00Z",
  "state": "requires_info",
  "question": "What is the unit age?",
  "field": "appliance_age",
  "urgency": "hard"
}
```

authNumber, callback callId, and callback eventId have maximum length 100. denyReason and question have maximum length 2000; field has maximum length 100. The failed.reason enum is busy, no_answer, ivr_error, agent_error, timeout, or aborted.

Event effects

Event	VRS Express ticket effect	Technician SMS
approved	Approve and complete using authNumber	Yes, normal approval SMS is attempted
partial	Approve and complete using authNumber and the partial detail	Yes, normal approval SMS is attempted
denied	Reject using denyReason	Yes, normal rejection SMS is attempted
failed	Status-only; no ticket decision	No
requires_info	Status-only; no ticket decision	No

Every accepted event is recorded and forwarded. SMS delivery is best effort, and replaying an event cannot repeat its ticket or SMS effect.

10. Callback acknowledgement, retry, and HMAC

VRS Express returns 200 after processing an event or recognizing an already processed event. Either result is delivery success. VRS AI retries network failures and callback HTTP 500, 502, or 503; unchanged 400 and 401 responses are not retryable.

The callback retry policy is a maximum of five attempts. Wait 1, 2, 4, and 8 seconds after successive failed attempts. Every retry keeps the same eventId and exact body bytes and refreshes only X-VRS-Timestamp.

Exact raw-byte HMAC

```
digest = HMAC-SHA256(callback_secret, exact_raw_request_body_bytes)
X-VRS-Signature = "sha256=" + lowercase_hex(digest)
```

- Do not parse and reserialize after signing.
- Do not include the timestamp, URL, or headers in the HMAC input.
- X-VRS-Event-Id must exactly equal body eventId.
- X-VRS-Timestamp is current Unix epoch milliseconds and must be within five minutes of VRS Express.
- Retries keep the same event ID and exact body but refresh the timestamp.

Deterministic test vector:

Raw body is one exact UTF-8 line; any visual wrapping is not a line break.

```
Secret: vrs-ai-test-secret-do-not-use-in-prod
Raw body: {"version":1,"eventId":"evt_001","callId":"vrs_ai_call_456","soNo":"13538277","occurredAt":"2026-07-
24T17:24:00Z","state":"approved","authNumber":"AHS-001","authorizedAmount":365.6}
Expected: sha256=5631d513c93853fa58efc382eed0c19db14ab42b94b1fa44938bb44ee352d4b6
```

11. Idempotency, forwarding, and terminal repeats

VRS Express uses a durable event ledger with separate apply and forward checkpoints. A replay of the same event returns 200 with deduped: true and cannot repeat the ticket decision or technician SMS. There is no time-based dedupe expiry in this interface.

requires_info may precede one terminal event. The terminal states are approved, partial, denied, and failed. VRS Express forwards the accepted event to <UPSTREAM_DEV_CALLBACK_URL> with its public call ID and the sender-provided callback secret.

TERMINAL REPEAT POLICY

Only active bridge calls return 409 DUPLICATE_CALL. V5 and nonretryable V8 terminal repeats remain subject to ordinary ticket and service-order policy. The narrow exception is a strict V8 request whose latest same-service-order bridge is a retryable failed V8 call with its terminal callback locally applied. VRS Express immediately creates a new normal ticket, new bridge row, and new public call ID. It excludes only prior retryable-failed V8 bridge tickets from legacy same-day duplicate checks; every other ticket constraint remains in force. It never reuses or resets the old ticket, bridge, or event. Upstream forwarding remains independent, so an old callback-forward retry may continue without delaying or altering the new request.

12. Exact development acceptance checklist

Direction A acceptance: VRS AI simulator to VRS Express

- [ ] Provision an active VRS Express DEV technician mapped to a valid four-digit district, then replace both sample tech_id occurrences.
- [ ] Replace each sample service_order with a unique eight-digit DEV value.
- [ ] Substitute <VRS_EXPRESS_DEV_BASE_URL> and <UPSTREAM_DEV_CALLBACK_URL> only in the development test runner.
- [ ] Send one unique complete V5 payload and receive 202 with a public call ID.
- [ ] Send one unique complete V8 payload and receive 202 with a public call ID.
- [ ] Confirm each request creates exactly one pending development ticket.
- [ ] Confirm VRS AI receives zero outbound requests before human approval.
- [ ] Confirm a partial V8 body returns 400 and does not fall through to V5.
- [ ] Confirm a duplicate active service order returns 409.

Direction B acceptance: VRS Express to VRS AI

- [ ] Substitute <VRS_AI_DEV_BASE_URL> and <VRS_EXPRESS_DEV_CALLBACK_URL> only in development configuration.
- [ ] Approve the V5 ticket and confirm exactly one strict V5 request arrives with only callback ownership replaced.
- [ ] Approve the V8 ticket and confirm exactly one strict V8 request arrives, with the same contract version and usable sender-signed photos.
- [ ] Return each of the five signed event shapes and confirm the expected ticket state, one technician SMS effect where applicable, and one signed upstream callback.
- [ ] Replay one event and confirm 200, deduped: true, and no repeated ticket or SMS effect.
- [ ] Send a retryable V8 failure, confirm its failed callback is applied locally, then immediately send a strict new V8 request and confirm a new normal ticket, new bridge row, and new public call ID. The old upstream forward retry may remain pending and must stay independent.
- [ ] Confirm V5 and nonretryable V8 terminal repeats remain subject to ordinary ticket and service-order policy.

PRODUCTION BOUNDARY

Do not use production URLs, credentials, or customer data for this checklist.

The included OpenAPI file is the implementation source for schemas and examples: vrs-express-vrs-ai-bridge-openapi.yaml.